

Deep Salient Detection for F0 Estimation

Changda Ma, Jiayi Wang

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1 Methodology

In this assignment, we replicate and evaluate a data-driven pitch tracking system using deep learning (Bitner, McFee, et al. 2017). The system learns a pitch salience representation from audio and decodes it into frame-wise fundamental frequency (F0) predictions. In addition to replicating the original fully convolutional model, we extended the evaluation by comparing three approaches: (1) a rule-based autocorrelation baseline, (2) the original CNN architecture, and (3) a transfer learning model based on a pretrained ResNet-18 network, which is the best accuracy model for the image classification task.

1.1 Model Architecture

We use the Harmonic Constant-Q Transform (HCQT) as the audio input representation, as it explicitly encodes the harmonic structure that is essential for pitch perception. All audio is resampled to 44.1 kHz, and we compute six Constant-Q transforms with harmonic scalings $h \in \{0.5, 1, 2, 3, 4, 5\}$. For each harmonic channel, we set the minimum frequency at $f_{\min} = C1 \approx 32.7Hz$, span six octaves, and use a resolution of 60 bins per octave, with a hop size of 512 samples. The CQTs of magnitude are stacked to form a tensor of shape $6 \times 360 \times T$ (harmonics $\times$ frequency bins $\times$ time frames). This multi-channel structure enables the model to capture spectral patterns associated with pitched sources across harmonics and sub-harmonics.

The target representation is a binary salience map aligned to the same frequency grid as the HCQT input. Ground-truth F0 annotations are quantized to frequency bins and Gaussian-blurred across nearby bins to reduce penalization for small frequency deviations and ensure stable learning during training, as shown in Figure 1.

Our primary model is a fully convolutional neural network that preserves time-frequency resolution throughout the architecture. Given an HCQT segment $\mathbf{x} \in \mathbb{R}^{H \times F \times T}$, the network consists of 3 stacked convolutional layers with batch normalization and

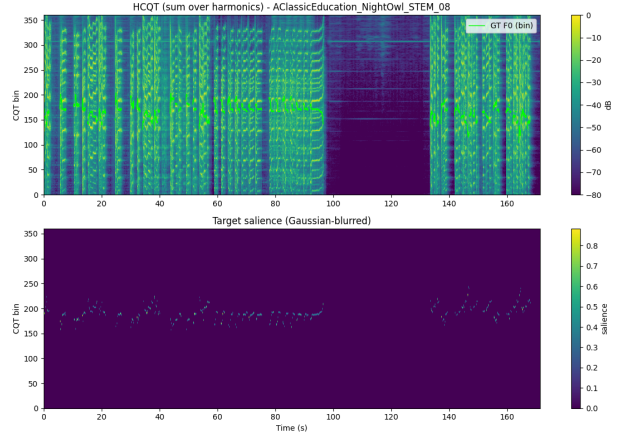


Figure 1: HCQT input representation and corresponding Gaussian-blurred F0 salience target of AClassicEducation NightOwl STEM 08.

ReLU activation, and intentionally omits pooling operations to maintain precise pitch localization. A final convolution 1×1 projects the hidden representation onto a single-channel salience map, producing logarithmic for each time-frequency bin. We refer to this architecture as “SimpleNet”. In addition, we train on randomly cropped 512-frame HCQT segments (batch size 8) using the Adam optimizer ($\text{lr} = 10^{-3}$). In the baseline, the model is optimized with a binary cross-entropy loss (BCE). In an experimental configuration, we adopt a hybrid objective that combines BCE on logits with an Mean Squared Error (MSE) term on the sigmoid outputs:

$$\mathcal{L}_{\text{hybrid}} = \lambda_{\text{BCE}} * \text{BCE} + \lambda_{\text{MSE}} * \text{MSE}$$

with $\lambda_{\text{BCE}} = 1.0$ and $\lambda_{\text{MSE}} = 0.1$. The BCE focuses on details of each bin decision, while the MSE encourages smoother alignment with the Gaussian-blurred target salience profiles in a global scene.

2 Experiment Setup

Conducted on Google Colab with an NVIDIA A100 high RAM GPU, we train and validate our models

using the MedleyDB-Pitch dataset, which provides frame-level fundamental frequency annotations for a musically diverse collection of vocal recordings. We create a custom artist-independent split, reserving approximately 20% of tracks for validation to avoid timbral overlap between training and evaluation data. During training, audio is processed into HCQT features and paired with salience target maps as described in the Methodology section.

We compare multiple architectures to explore the effects of model complexity and loss design. First, we use “SimpleNet”, a lightweight fully convolutional architecture introduced in the methodology part, and train it under two objectives, including BCE, and a hybrid BCE+MSE loss to compare with. Second, we also conducted experiments based on ResNet18 pretrained on ImageNet. We study both a frozen backbone and a fine-tuned configuration that updates all layers with a reduced learning rate ($lr=1e-4$). The reason why we reduce learning rate is that the ResNet18 uses pretrained ImageNet weights, which require more conservative updates to avoid overwriting learned representations. For each configuration, we similarly test the BCE-only loss and the BCE+MSE hybrid to access whether smoothness based regularization benefits large pretrained models. In total, we have six models, which are shown in the Table 1.

ID	Backbone	Pretrained	Frozen	Loss	LR
M1	SimpleNet	No	N/A	BCE	10^{-3}
M2	SimpleNet	No	N/A	BCE + MSE	10^{-3}
M3	ResNet-18	Yes	Yes	BCE	10^{-4}
M4	ResNet-18	Yes	Yes	BCE + MSE	10^{-4}
M5	ResNet-18	Yes	No	BCE	10^{-4}
M6	ResNet-18	Yes	No	BCE + MSE	10^{-4}

Table 1: Model configurations used in our experiments.

During training, we visualize the loss to examine the convergence as shown in Figure 2. “SimpleNet” models show fast and stable convergence. ResNet models start with higher loss due to domain mismatch, with fine-tuned variants converging more effectively than frozen ones. Hybrid loss models show smoother learning dynamics, suggesting that the MSE term regularizes BCE and encourages globally coherent salience predictions. Overall, the curves show good convergence and clear differences in optimization behavior between models.

We evaluate model performance using the Voca-dito evaluation dataset, which contains professionally annotated singing-voice melodies. This ensures an unbiased test scenario, since these tracks are unseen during training and contain different recording conditions and singers.

To compare different system architectures, we train

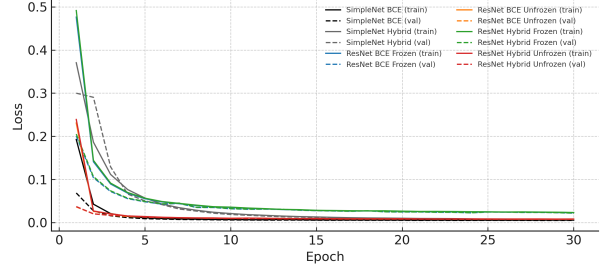


Figure 2: Six Models Train/Val Loss.

and evaluate three pitch tracking approaches: (1) a rule-based autocorrelation baseline, implemented from scratch, (2) the replicated fully convolutional CNN model Bittner, McFee, et al. 2017, and (3) a transfer learning system based on ResNet18 pretrained on ImageNet. The baseline operates directly on time-domain audio, selecting periodicity peaks per frame, while the deep learning systems predict salience maps that are decoded to F0 contours by selecting the maximum-salience frequency bin. To ensure a fair comparison, we performed a threshold sweep for the salience detection output. We selected the threshold value that maximized Overall Accuracy (OA) and applied this optimized setting for the final evaluation on the test set.

2.1 Evaluation Metrics

We evaluate all systems using the standard melody metrics provided by the `mir_eval.melody` module (Bittner and Bosch 2019).

- **Voicing Recall (VR):** Percentage of frames where the system correctly detects a voiced melody when one is present.
- **Voicing False Alarm (VFA):** Percentage of frames where the system predicts voicing when the reference is unvoiced.
- **Raw Pitch Accuracy (RPA):** Fraction of correctly estimated fundamental frequencies (within 50 cents), evaluated only on voiced frames.
- **Raw Chroma Accuracy (RCA):** Variant of RPA that ignores octave errors by folding estimates into a single octave.
- **Overall Accuracy (OA):** Combined measure reflecting correct pitch estimates on voiced frames and correct voicing decisions on unvoiced frames; used as the primary score for system comparison.

Model	VR $\uparrow$	VFA $\downarrow$	RPA $\uparrow$	RCA $\uparrow$	OA $\uparrow$
Autocorrelation	99.85 $\pm$ 0.17	70.12 $\pm$ 9.38	84.09 $\pm$ 7.56	84.72 $\pm$ 6.79	65.65 $\pm$ 7.49
SimpleNet (BCE)	74.92 $\pm$ 13.30	1.54 $\pm$ 1.50	72.71 $\pm$ 13.21	72.74 $\pm$ 13.19	81.58 $\pm$ 9.03
SimpleNet (BCE+MSE)	72.18 $\pm$ 16.55	3.18 $\pm$ 3.47	69.47 $\pm$ 15.41	69.49 $\pm$ 15.39	78.95 $\pm$ 9.95
ResNet18 (BCE Frozen)	3.43 $\pm$ 3.88	10.30 $\pm$ 8.64	0.00 $\pm$ 0.00	0.15 $\pm$ 0.35	30.33 $\pm$ 6.44
ResNet18 (BCE Unfrozen)	55.03 $\pm$ 11.45	23.33 $\pm$ 9.35	12.93 $\pm$ 5.94	12.93 $\pm$ 5.94	34.81 $\pm$ 6.75
ResNet18 (BCE+MSE Frozen)	3.43 $\pm$ 3.88	10.32 $\pm$ 8.67	0.00 $\pm$ 0.00	0.15 $\pm$ 0.35	30.32 $\pm$ 6.44
ResNet18 (BCE+MSE Unfrozen)	54.85 $\pm$ 11.67	23.56 $\pm$ 9.54	13.01 $\pm$ 6.17	13.01 $\pm$ 6.17	34.78 $\pm$ 6.92

Table 2: Melody extraction performance on the Vocabito dataset (%). Mean $\pm$ standard deviation.

3 Results

Table 2 summarizes the melody extraction performance on the Vocabito dataset for all evaluated architectures. The rule-based Autocorrelation baseline achieves near-perfect Voicing Recall (99.85%), but this comes at the cost of a high Voicing False Alarm rate (70.12%), which compromises its Overall Accuracy. In contrast, the replicated CNN-based SimpleNet models yield the most robust performance, achieving an Overall Accuracy of 81.58% with fewer false alarms (1.54%).

The transfer-learning approaches using ResNet18 underperform across all metrics. When the ImageNet-pretrained layers remain frozen, the model fails to generate meaningful pitch estimates (RPA $\approx$ 0%). While unfreezing the layers yields a slight improvement in pitch sensitivity, the performance remains significantly below that of the SimpleNet models. We attribute this deficiency to the domain mismatch between the ResNet18 model and the specific spectral structure of HCQT audio representations.

Overall, these results suggest that the convolutional neural network for salient detection is better suited for F0 estimation than the autocorrelation method. While Autocorrelation detects periodicity effectively, its inability to reject unvoiced frames limits its reliability for real-world melody estimation.

3.1 Discussion

The stark performance gap between the SimpleNet and ResNet18 architectures highlights a fundamental challenge in applying computer vision paradigms to audio signal processing. While ResNet18 is a powerful feature extractor for object recognition, its pretraining on ImageNet is inherently misaligned with the task of melody salience detection.

Image classification models are designed to learn translation-invariant features (e.g., recognizing a shape regardless of its location). However, in the time-frequency domain, the vertical axis represents

frequency; therefore, "objects" (harmonic stacks) are not translation invariant—shifting a pattern vertically changes its pitch and musical meaning. The filters learned by ResNet to detect edges and textures in natural images do not generalize well to salient detection.

Furthermore, our results show that even unfrozen fine-tuning could not bridge this gap. This suggests that the deep architectural priors of the ResNet model, combined with initialization weights biased toward visual classification, hindered the model’s ability to converge on the correct spectral features. In contrast, SimpleNet, trained from scratch with a lightweight architecture, was able to learn task-specific features directly from the audio data without the interference of conflicting inductive biases. This confirms that a smaller, domain-specific model often outperforms a larger, generalized vision model for dense prediction tasks like spectral salience.

References

- Bittner, Rachel, Brian McFee, et al. (Oct. 2017). “Deep Salience Representations for F0 Estimation in Polyphonic Music”. In: *Proceedings of the 18th International Society for Music Information Retrieval Conference (ISMIR)*. Suzhou, China, pp. 63–70.
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